Project Design Phase-II: Solution Requirements (Functional & Non-functional)

Date:	30 September 2026
Team ID:	[SWTID-2026-3230]
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	4 Marks

Functional Requirements

Following are the functional requirements of the proposed Auto Ticket Classification using Flow Designer solution:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Portal Access	Login via Single Sign-On (SSO / University Credentials) Role-based access control for Students, Faculty, Agents, and Admins
FR-2	Ticket Submission	Submit IT support ticket via Service Portal Input fields for Short Description, Detailed Description, and Attachment upload
FR-3	Automated Flow Designer Triage	Trigger Flow Designer workflow on new incident record creation Keyword extraction and pattern matching against Short Description / Description Automated assignment of Category, Subcategory, and Priority

FR-4	Smart Routing & Notification	Automatic routing of classified tickets to respective Assignment Groups (e.g., Network, Hardware, Software) Instant email notification sent to the end user and assigned support team
FR-5	Queue & Exception Management	Support executive dashboard for viewing pre-classified queue Manual override and re-classification capability for edge-case tickets with audit trail

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	**Non-Functional Requirement**	**Description**
NFR-1	Usability	The ServiceNow Service Portal and dashboard must provide an intuitive, responsive, and accessible user interface for students, faculty, and support agents across desktop and mobile devices.
NFR-2	Security	Ensure strict role-based access control (RBAC), data encryption in transit (HTTPS/TLS) and at rest, and secure API credentials adhering to university IT security standards.
NFR-3	Reliability	The Flow Designer classification workflows must execute reliably without data loss or pipeline failures, with

		robust error-handling and fallback default routing for ambiguous descriptions.
NFR-4	Performance	Flow Designer automated ticket classification and routing must complete within 3 seconds of ticket submission to ensure near real-time queue assignment.
NFR-5	Availability	The system must maintain 99.9% uptime during operational academic hours, leveraging ServiceNow cloud infrastructure redundancy.
NFR-6	Scalability	The architecture must seamlessly handle peak ticket volumes during semester beginnings and exam periods without degradation in classification speed or system responsiveness.